

CAP5415

Computer Vision

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HEC-241

Autoencoder

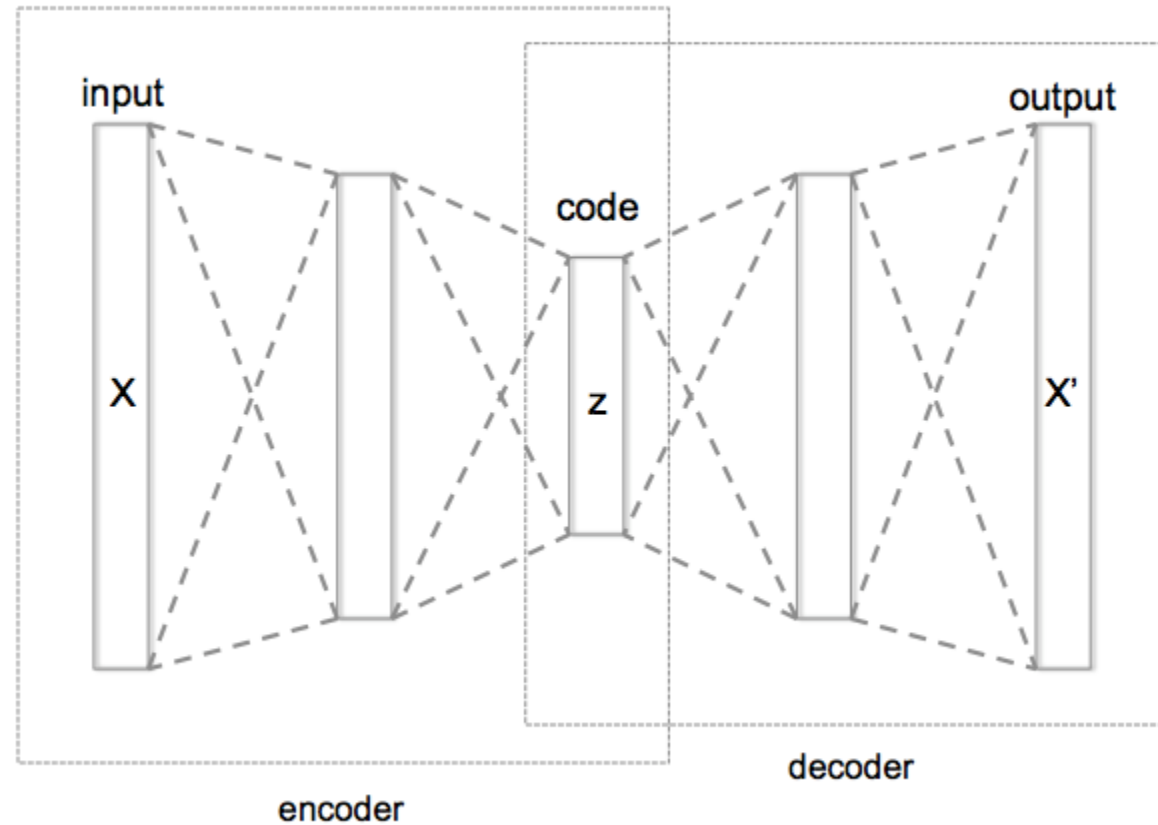
Lecture 10

Autoencoder

- Reproduce the input
 - Via learning features
- Unsupervised learning
 - Efficient way to learn features
 - Still need a loss function – implicit supervision
- Supervised learning
 - Need labels/annotations

Autoencoder

- Encoder – decoder
- Encoding
 - Key idea



Autoencoder

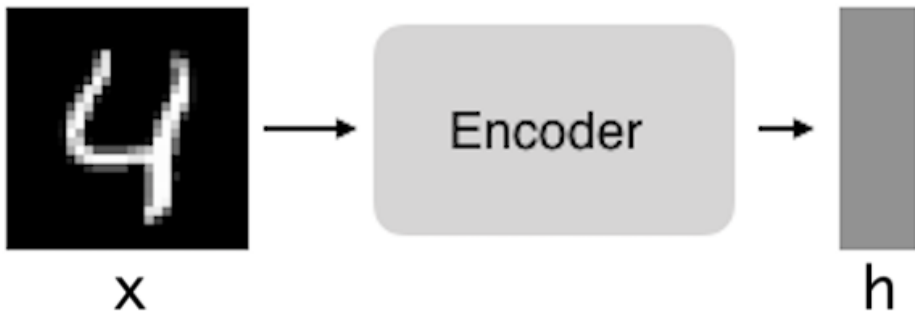
- Compare PCA/SVD
 - PCA produce smaller set of vectors
 - Approximate the input vectors via linear combination.
 - Very efficient for certain applications.
- Autoencoder
 - Can learn nonlinear dependencies
 - Can use convolutional layers
 - Can use transfer learning

Autoencoder

- Encoder: $h = f(x)$
 - Compress input into a latent-space
 - Usually smaller dimension
- Decoder: $r = g(f(x))$
 - Reconstruct input from the latent space

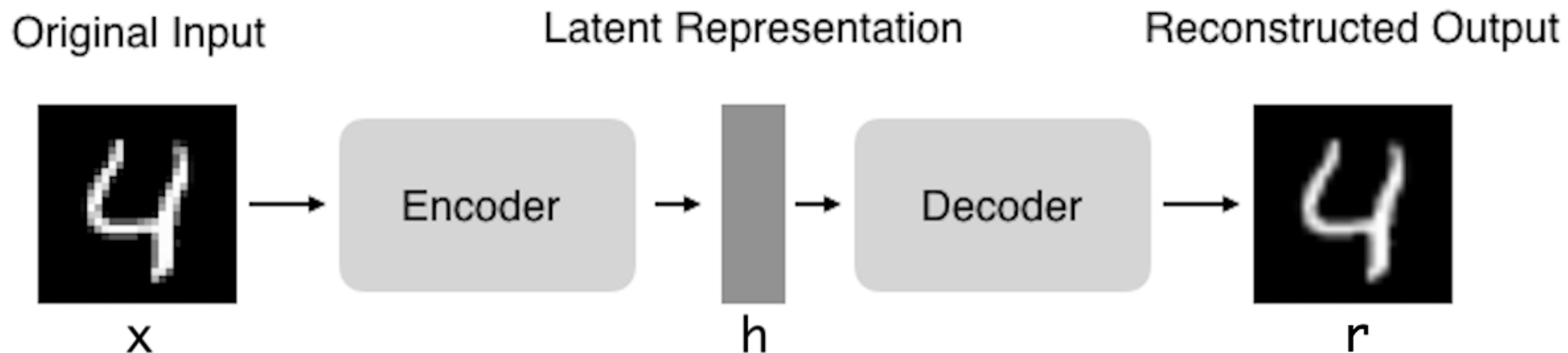
Original Input

Latent Representation



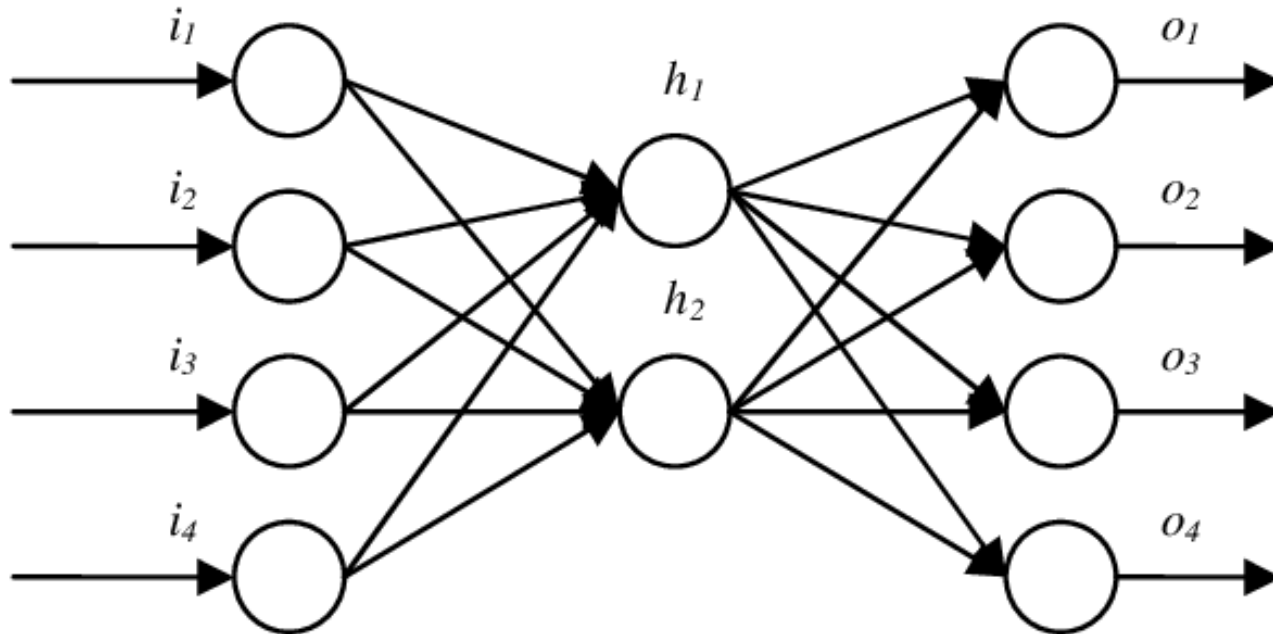
Autoencoder

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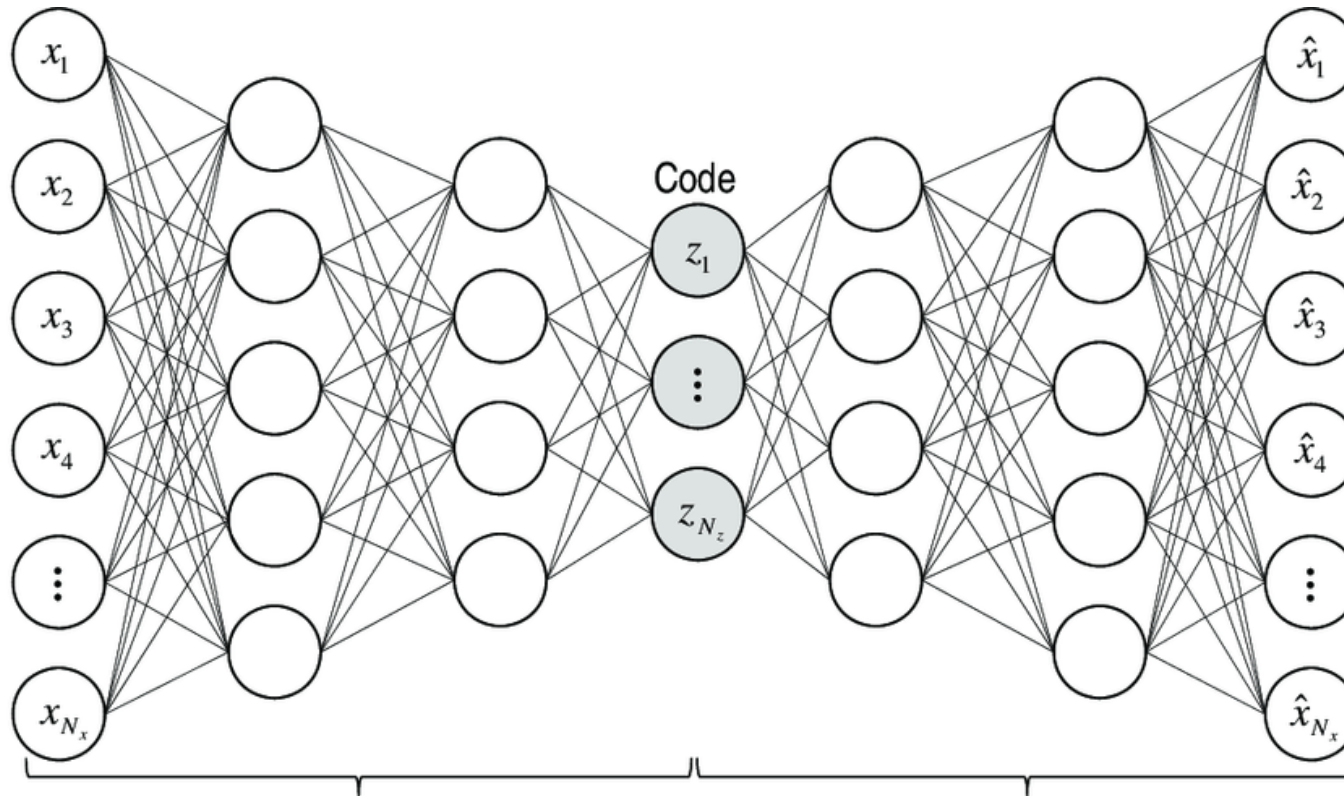
Autoencoder

- Shallow



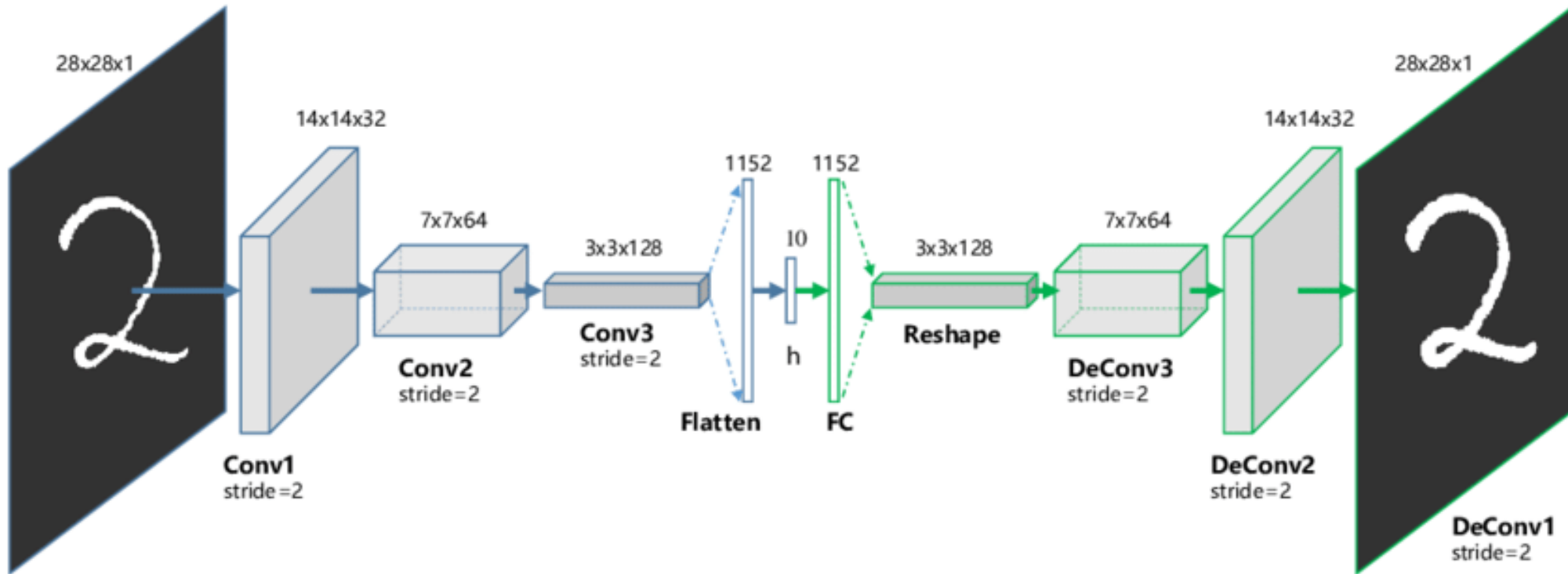
Autoencoder

- Deep



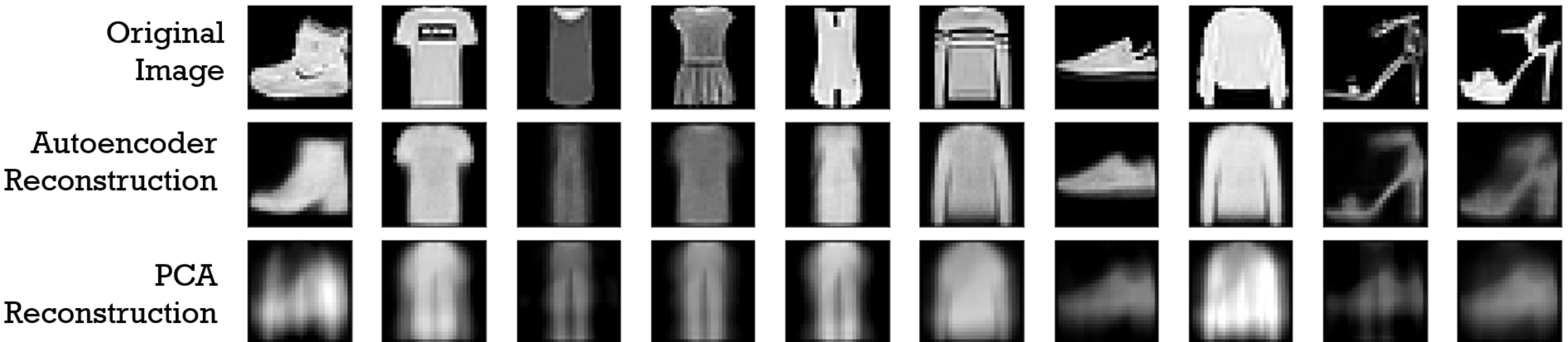
Autoencoder

- CNN



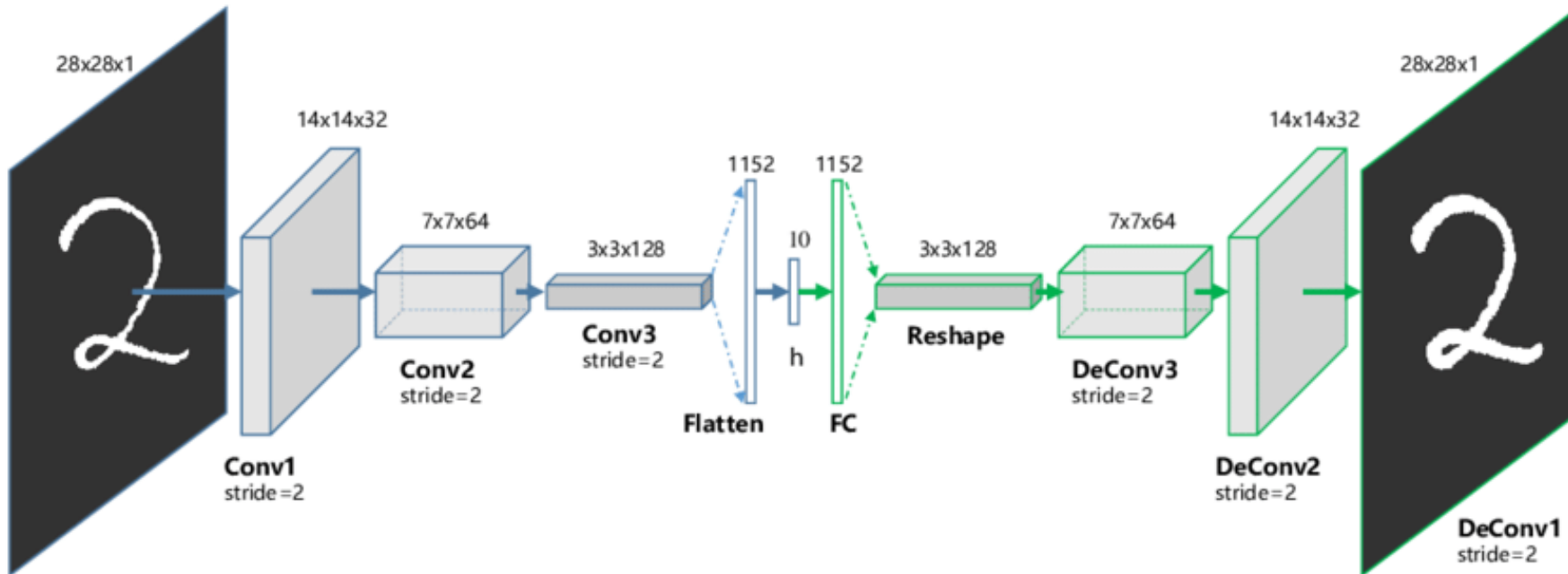
Autoencoder

- Reconstruction
 - Latent vector of size 2
 - Compression from 28x28



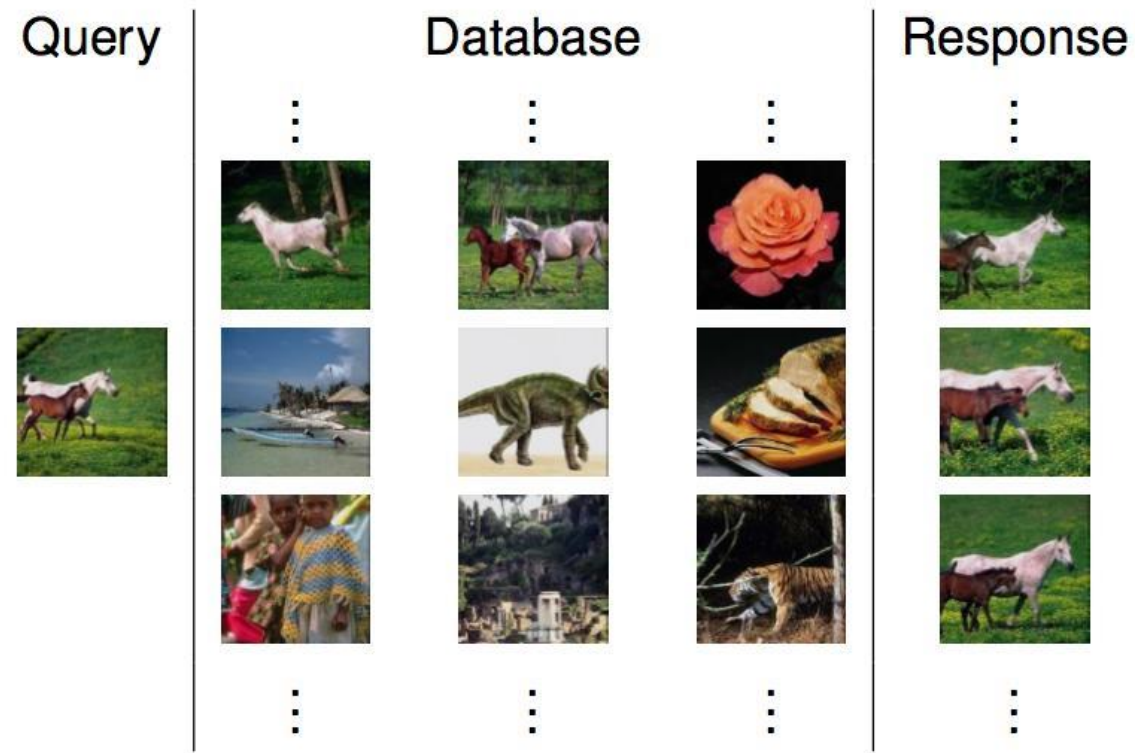
Feature learning

- Define a loss function
 - MSE, CE, etc.
- Optimize



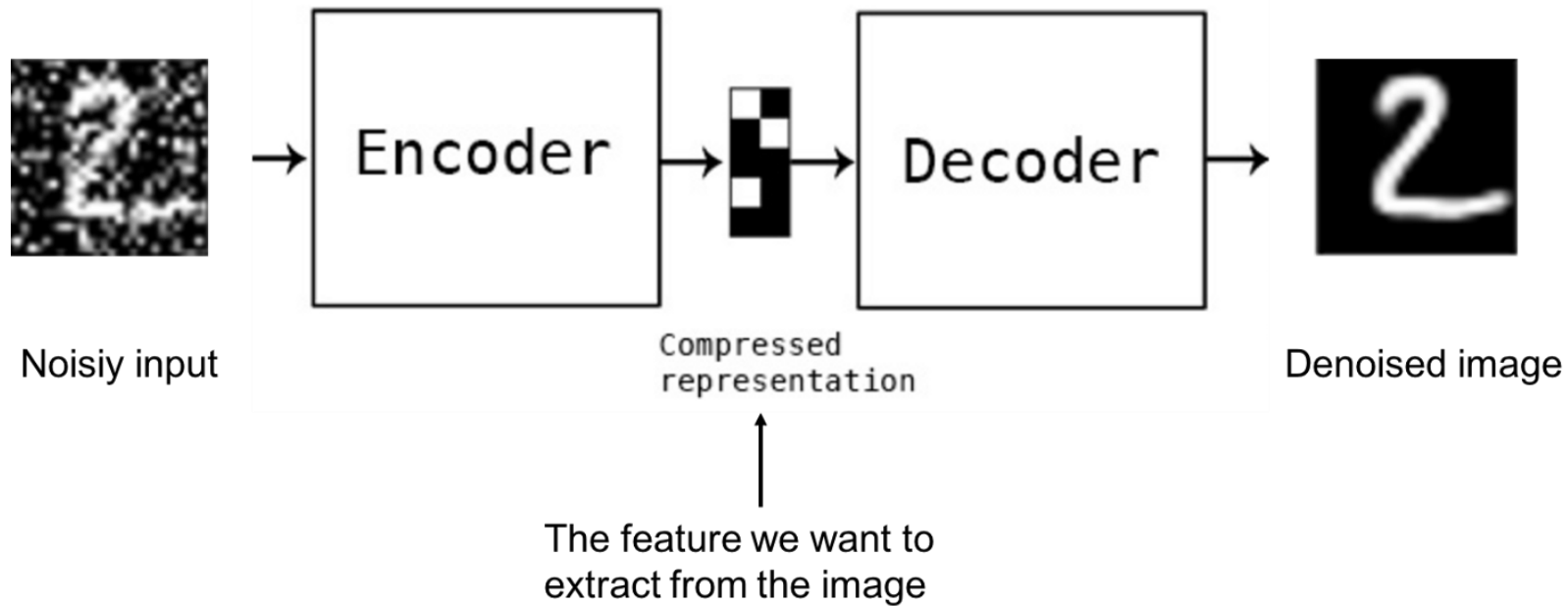
Feature learning

- Image retrieval
 - Dimensionality reduction helps



Autoencoder – application

- Denoising



Autoencoder – application

- Image colorization



Autoencoder – application

- Anomaly detection



Properties

- Data-specific
 - Compress data similar to what they have been trained on
- Lossy
 - Outputs will be degraded compared to the original inputs
- Learned automatically from examples
 - It is easy to train
 - It will perform well on data similar to training samples
- Compare with hand-crafted features

Questions?